



ECCV 2026 | MALMÖ | SEPT 8-12



北京大学



DeepImmersion



鹏城实验室
PENGCHENG LABORATORY

Online Segment 3D Gaussians via Launching Virtual Drones

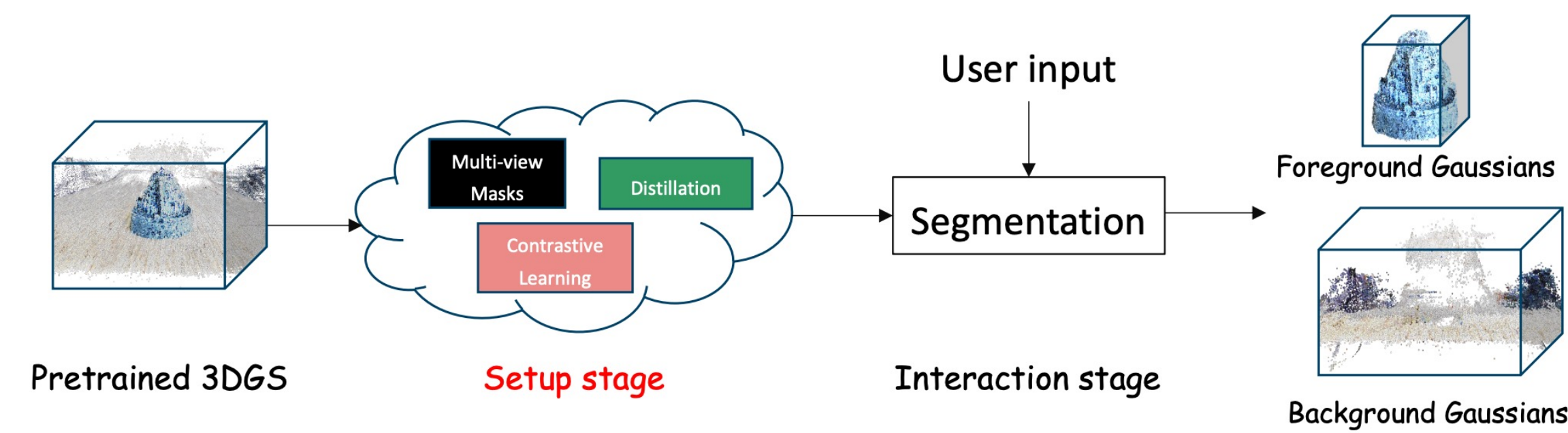
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Question: What Problem Do We Solve?



- Most existing 3DGS segmentation methods operate offline;
- The setup stage is time-consuming (often tens of minutes);
- Existing online methods require minute-level inference.

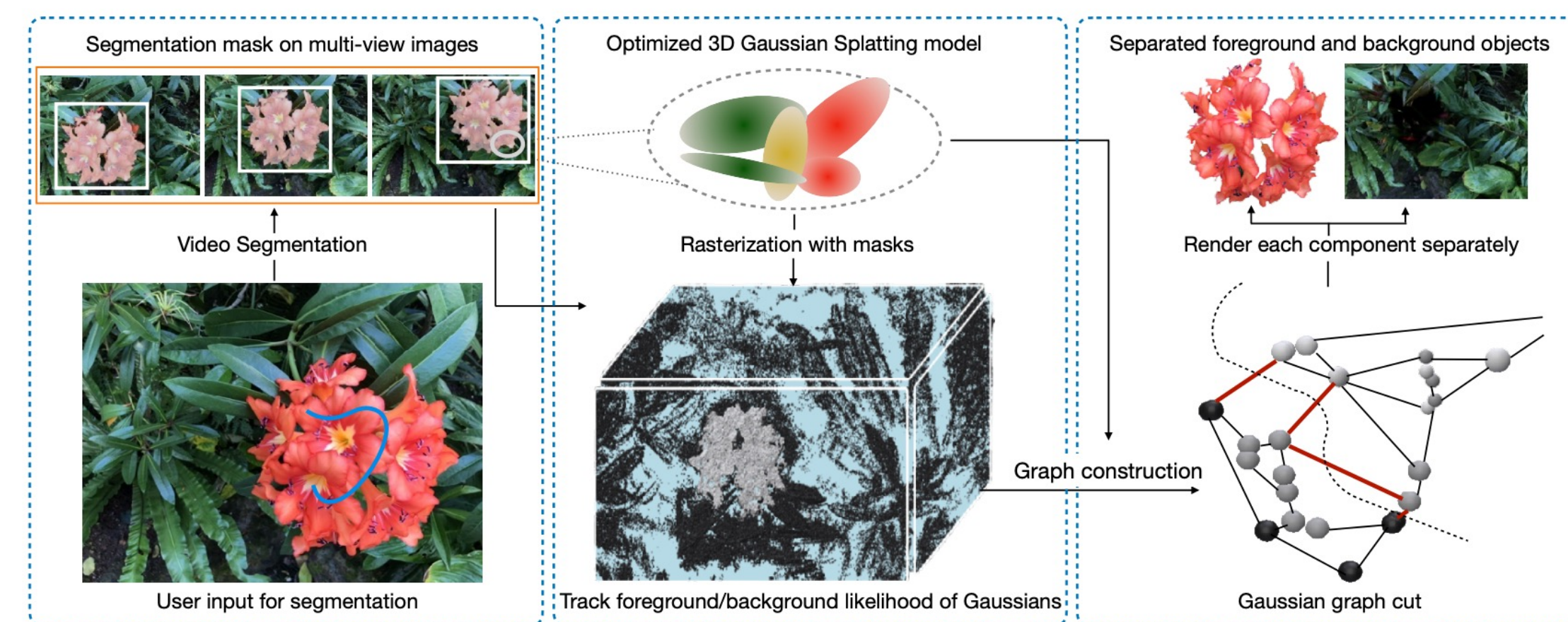


Existing methods trade off setup time against segmentation time.

Answer: We Eliminate the Setup Stage!

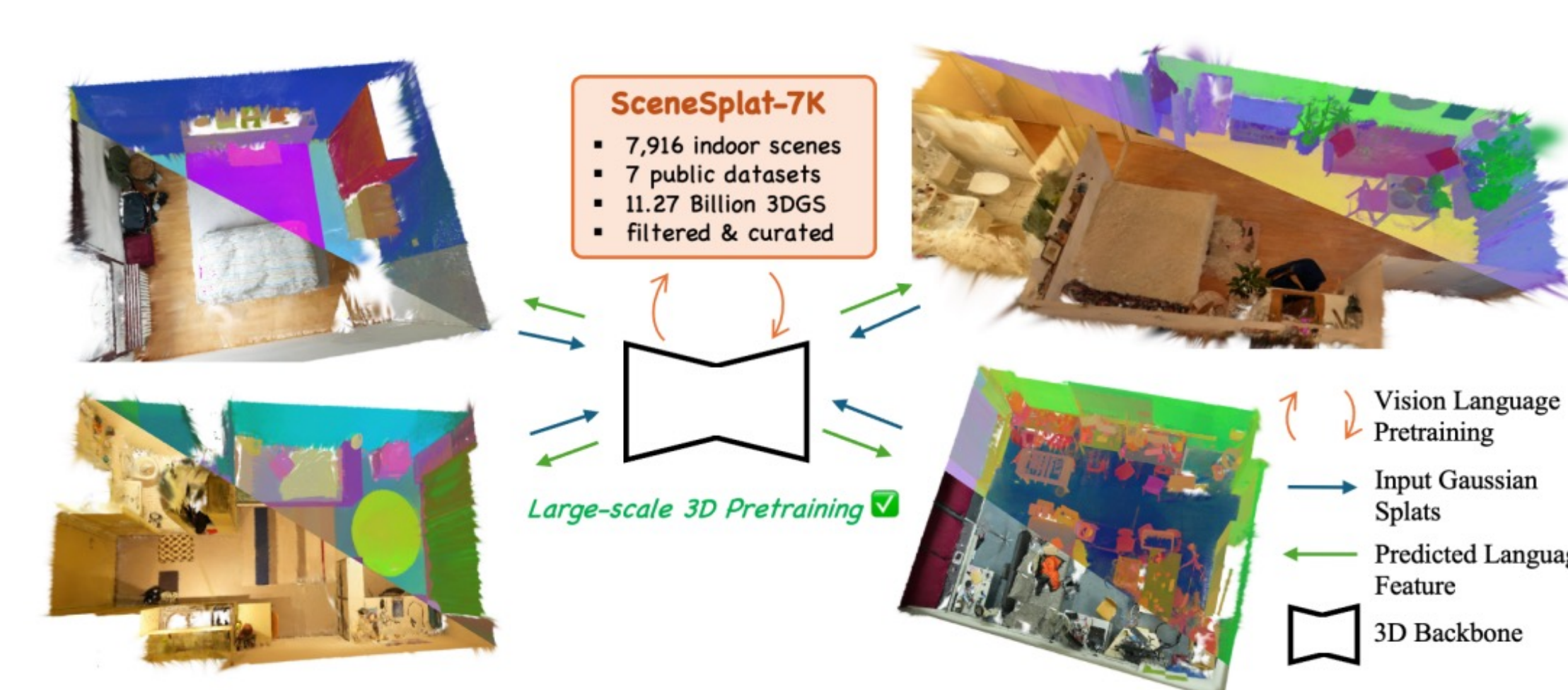
The challenge of eliminating setup stage

1. GaussianCut – Online Segmentation



Online mask propagation and graph construction require minute-level segmentation!

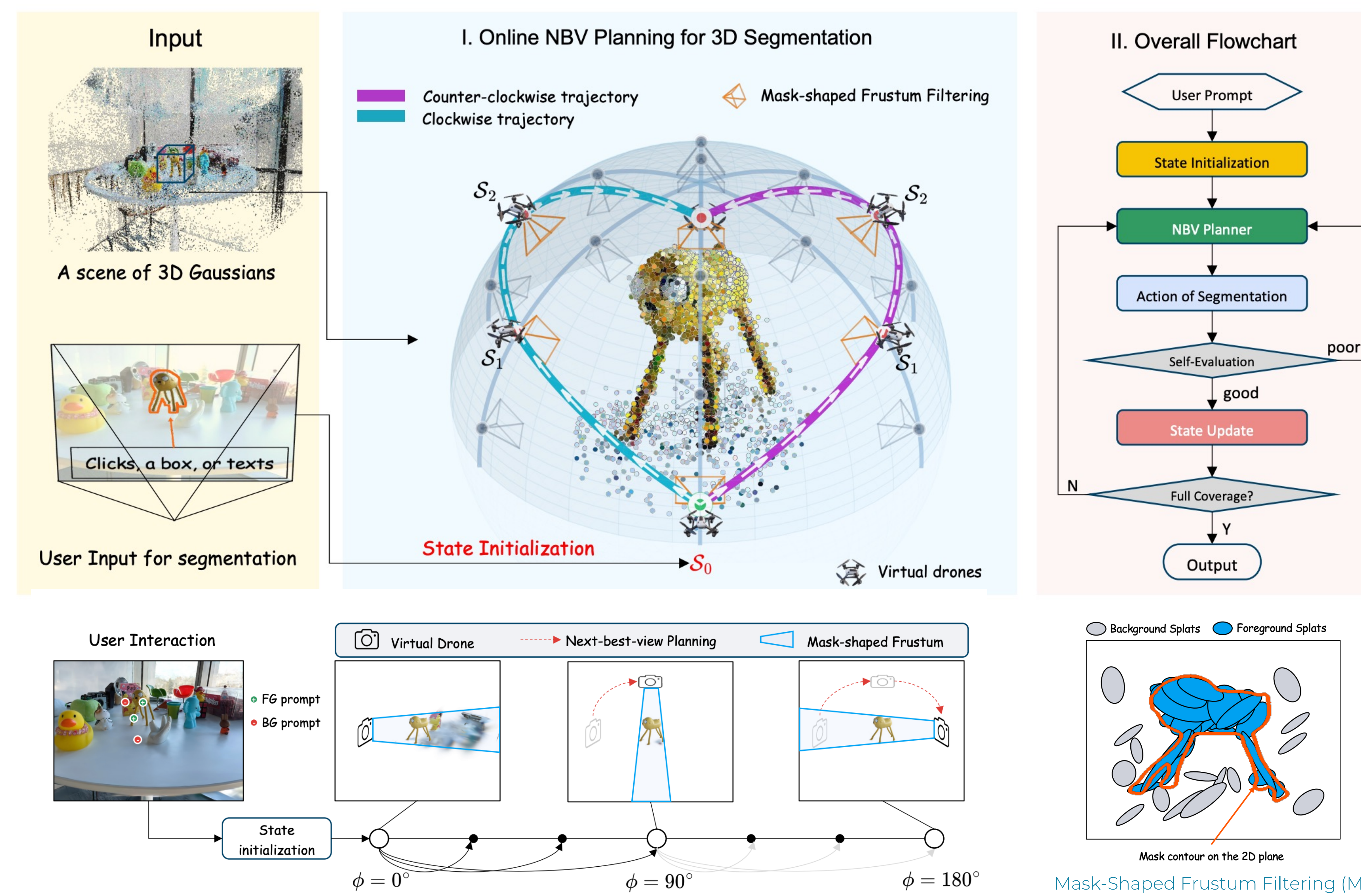
2. SceneSplat – 3D Pretraining



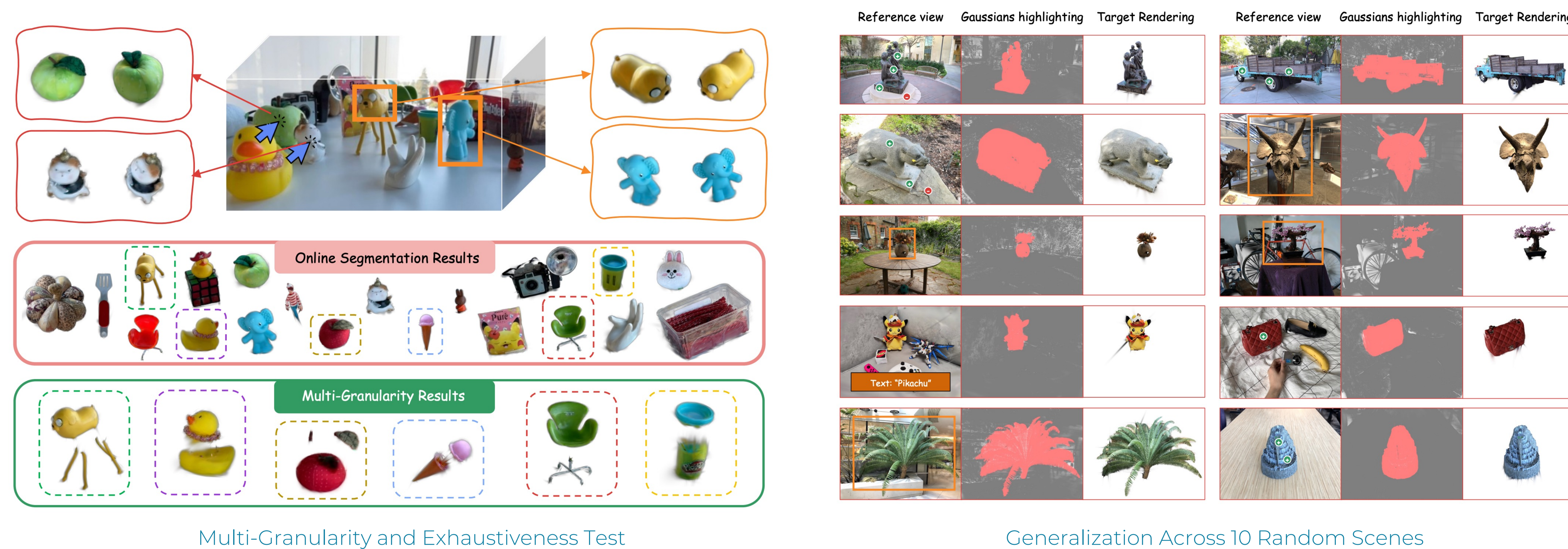
SceneSplat (3D)	SAM (2D)
7k samples	11,000k samples

Existing 3D pretraining data are too sparse to achieve SAM-level generalization.

Our Method: SAGO -- Segment Any 3D Gaussians Online



Qualitative Results



Quantitative Results

Method	Type	SPIn-NeRF		NVOS		Time	
		mIoU	mAcc	mIoU	mAcc	Setup	Seg.
MVSeg [30]	Offline	90.4	98.8	-	-	-	-
NVOS [35]	Offline	-	-	70.1	92.0	-	-
ISRF [10]	Offline	71.5	95.5	83.8	96.4	-	-
SA3D [5]	Offline	91.9	98.8	90.3	98.2	2~5m	15~30s
LangSplat [32]	Offline	69.5	94.5	74.0	94.0	~2.5h	-
SAGA [4]	Offline	93.4	99.2	92.6	98.6	~1h	10ms
FlashSplat [37]	Online	-	-	91.8	98.6	30~60s	1~2s
iSegMan [53]	Online	92.4	99.1	92.0	98.4	50s~90s	4~6s
GaussianCut [12]	Online	92.9	99.2	92.5	98.4	N/A	40~60s
Ours	Online	92.5	99.3	92.7	98.7	N/A	0.4~0.9s

- SAGO eliminates setup while preserving tier-1 performance!

- Over 50x faster than prior setup-free work: 50 s -> 1 s

Table 2: Comparison of interactive 3D segmentation on SPIn-NeRF [30] and NVOS [35]. First, second, and third best results are highlighted.

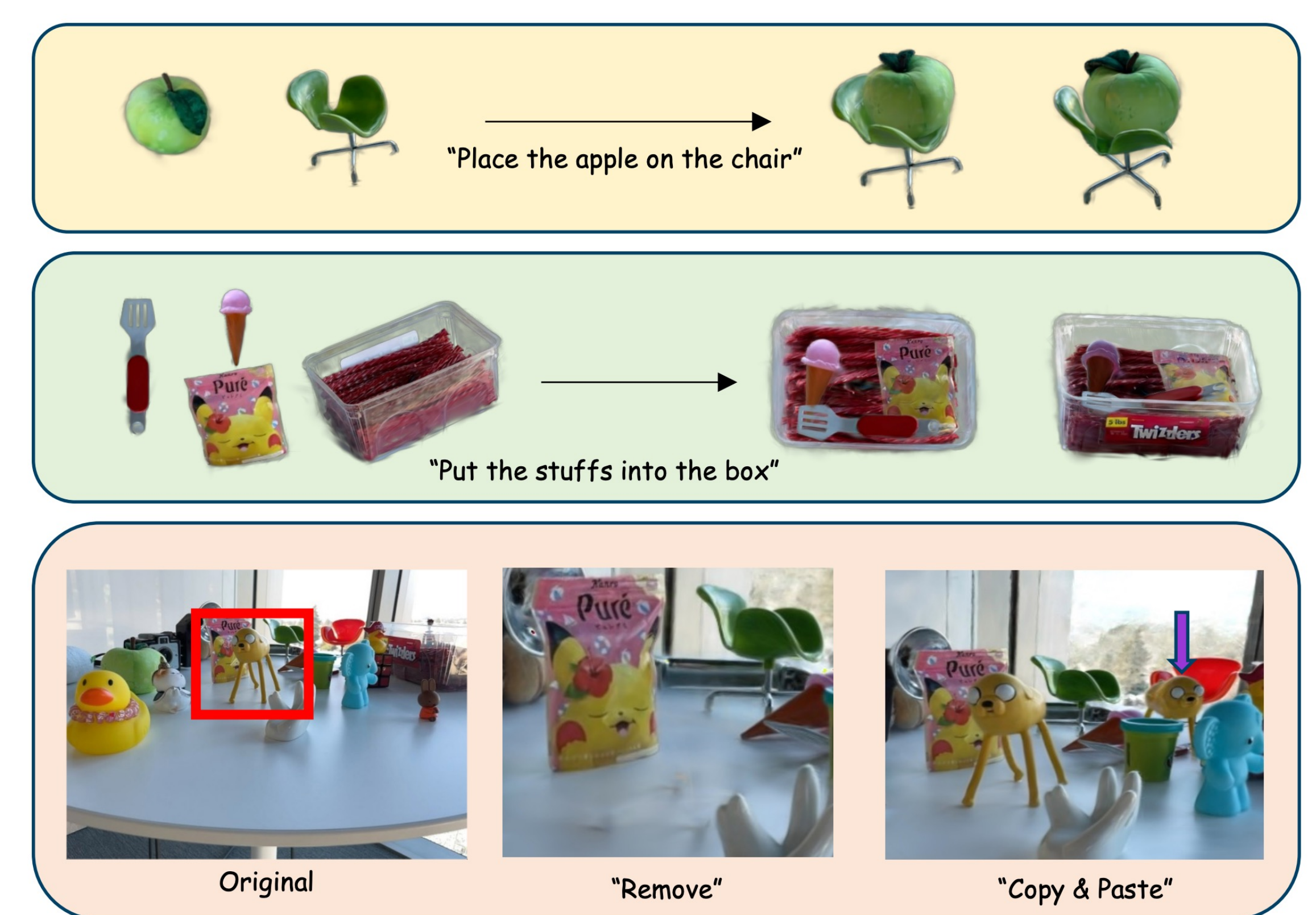
Table 3: Quantitative comparison on LERF-Mask (mIoU $\uparrow$).

Method	Figurines	Ramen	Teatime	Average
Lerf [15]	33.5	28.3	49.7	37.1
Gau-Group [48]	69.7	77.0	71.7	66.1
OmniSeg3D [49]	87.1	77.3	73.8	79.4
ClickGaussian [6]	93.2	90.9	83.2	89.1
Ours	93.9	90.6	88.6	91.0

Table 4: Quantitative comparison on 3D-OVS dataset (mIoU $\uparrow$).

Method	bed	bench	room	sofa	lawn	mean
LERF [15]	73.5	53.2	46.6	27.0	73.7	54.8
3D-OVS [25]	89.5	89.3	92.8	74.0	88.2	86.8
LangSplat [32]	92.5	94.2	94.1	90.0	96.1	93.4
N2F2 [2]	93.8	92.6	93.5	92.1	96.3	93.9
SAGA [4]	97.4	95.4	96.8	93.5	96.6	96.0
Ours	98.5	96.6	95.7	94.3	95.9	96.2

Online Editing



Takeaways

- Virtual drones enable fast, cost-effective 3D task execution.
- SAGO also serves as a 3D annotator for 3D pretraining, labeling an object in only 500 ms.
- SAGO can be readily extended to 4D.

Contacts



Video Demo

